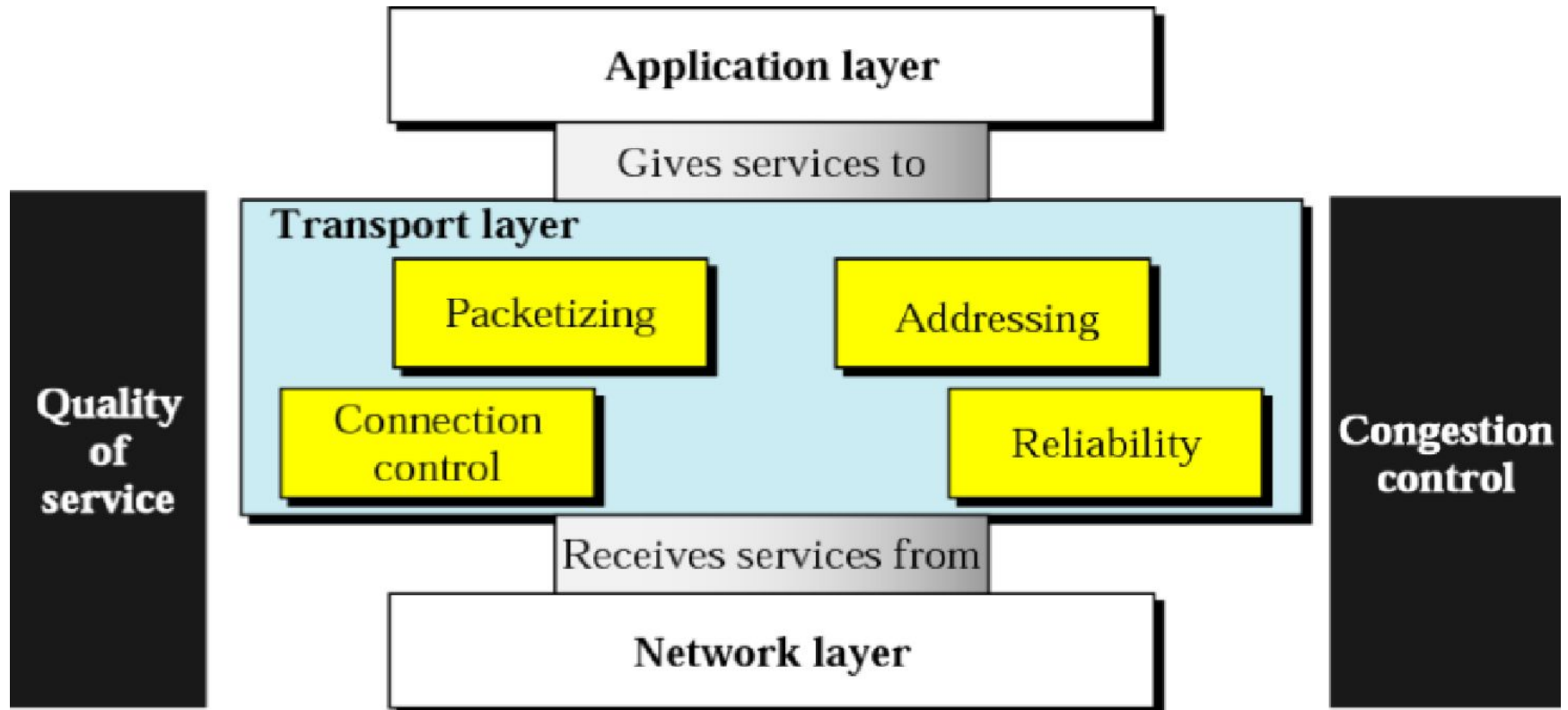


Data Communication & Networks

Lecture 27
Transport layer
UDP, TCP

Transport Layer



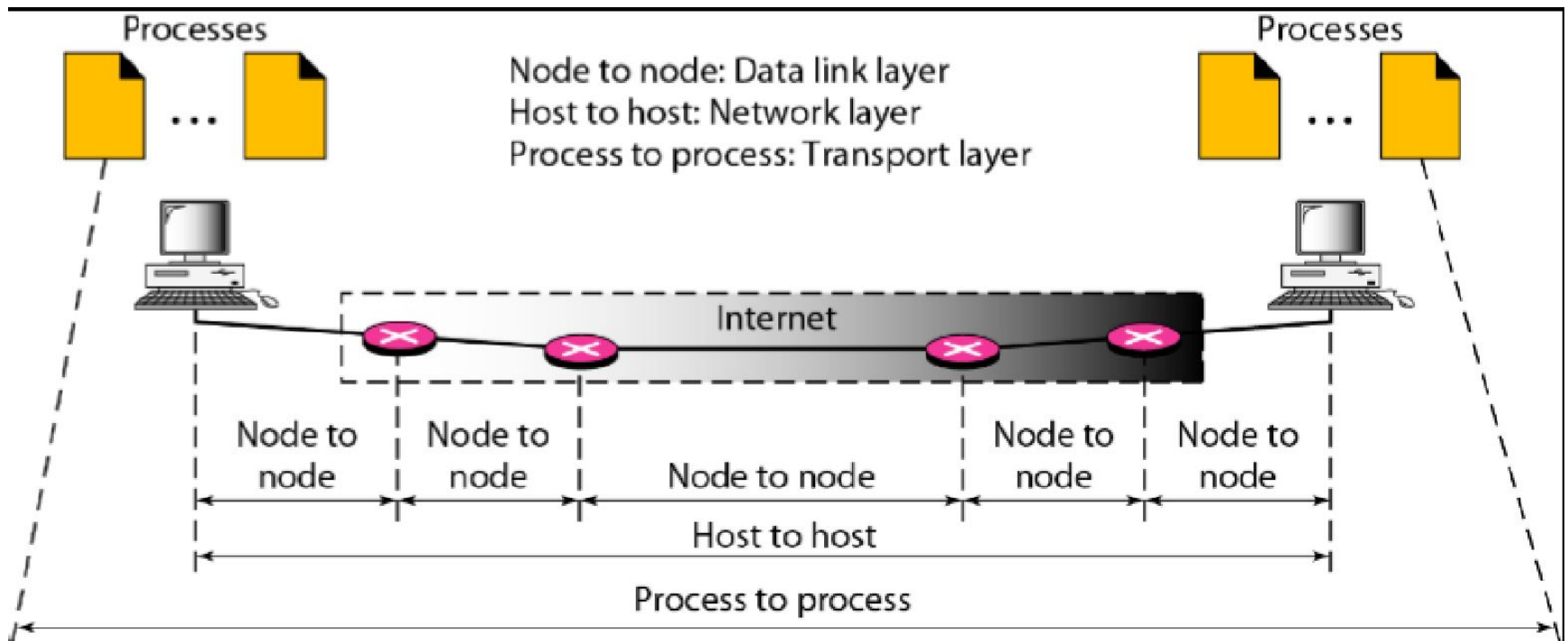
Transport Layer Topics to Cover

- ***UDP, TCP***

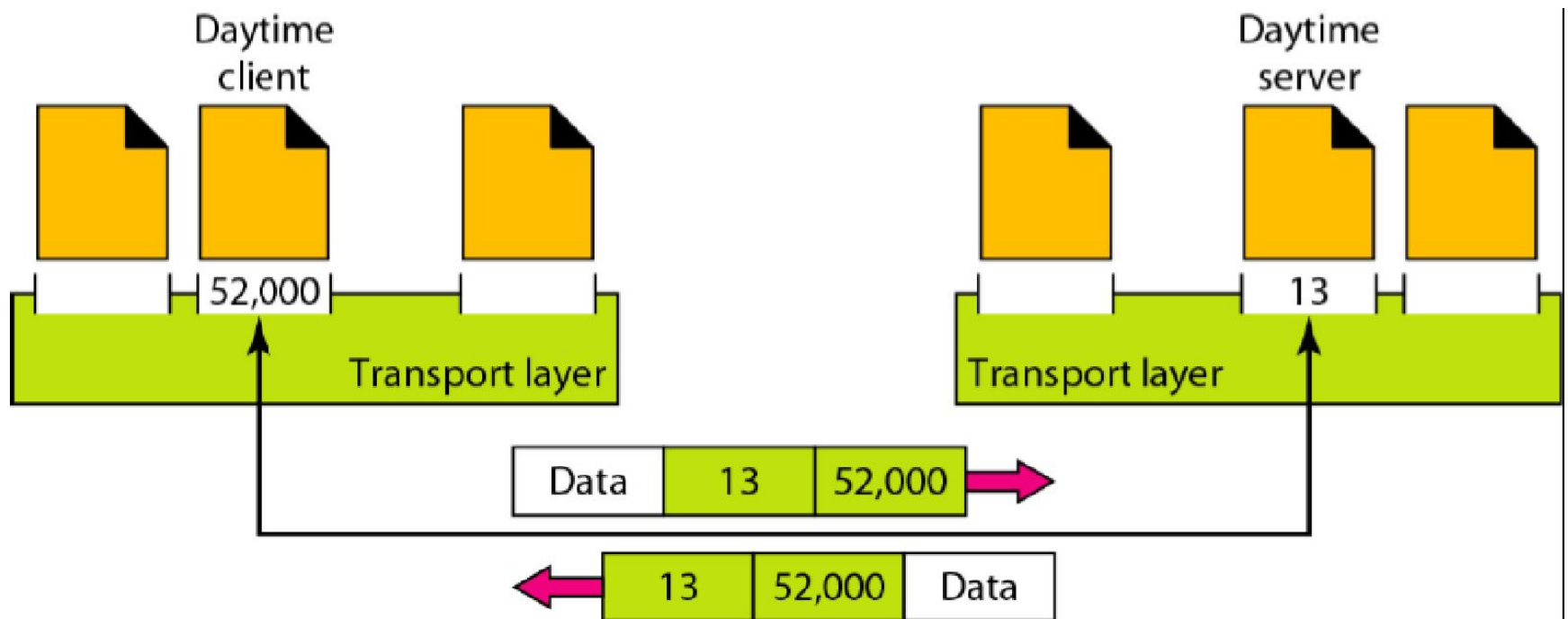
Process-to-Process Delivery

- The transport layer is responsible for
- process-to-process delivery—the delivery of
- a packet, part of a message, from one
- process to another. Two processes
- communicate in a client/server relationship,
- as we will see later.

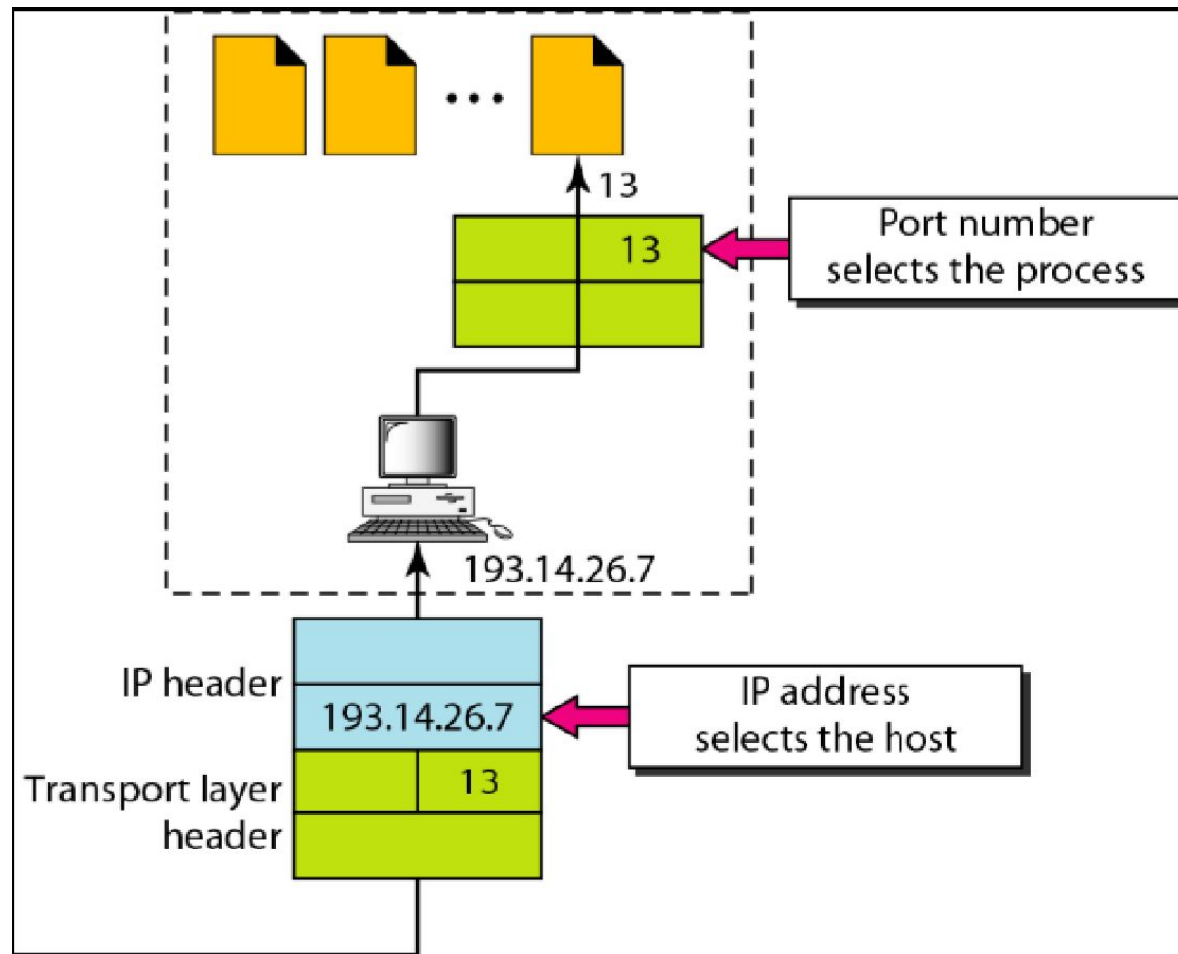
Types of Data Deliveries



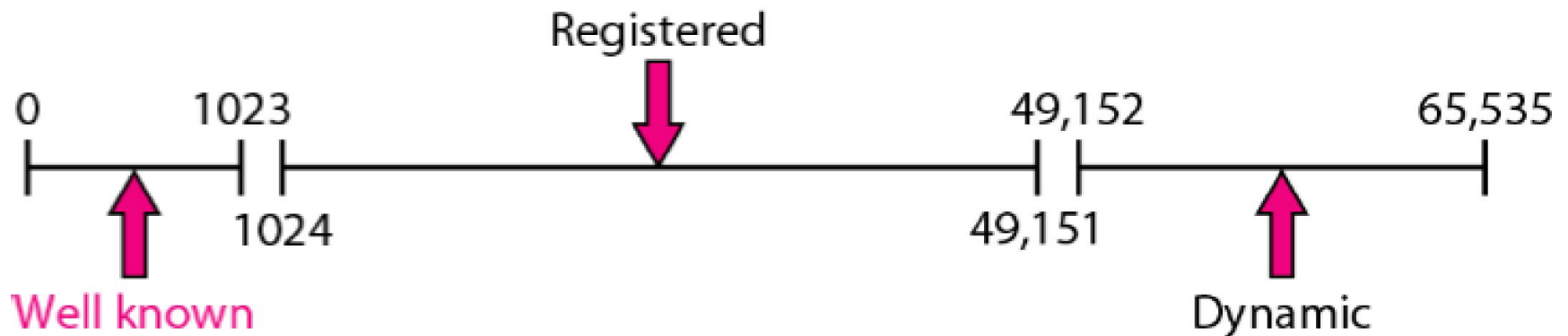
Port Numbers



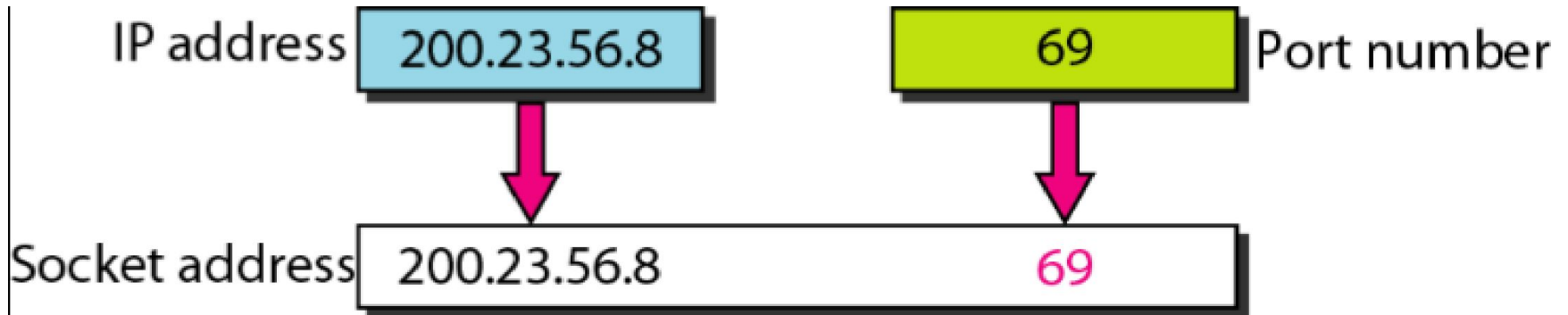
IP addresses versus port numbers



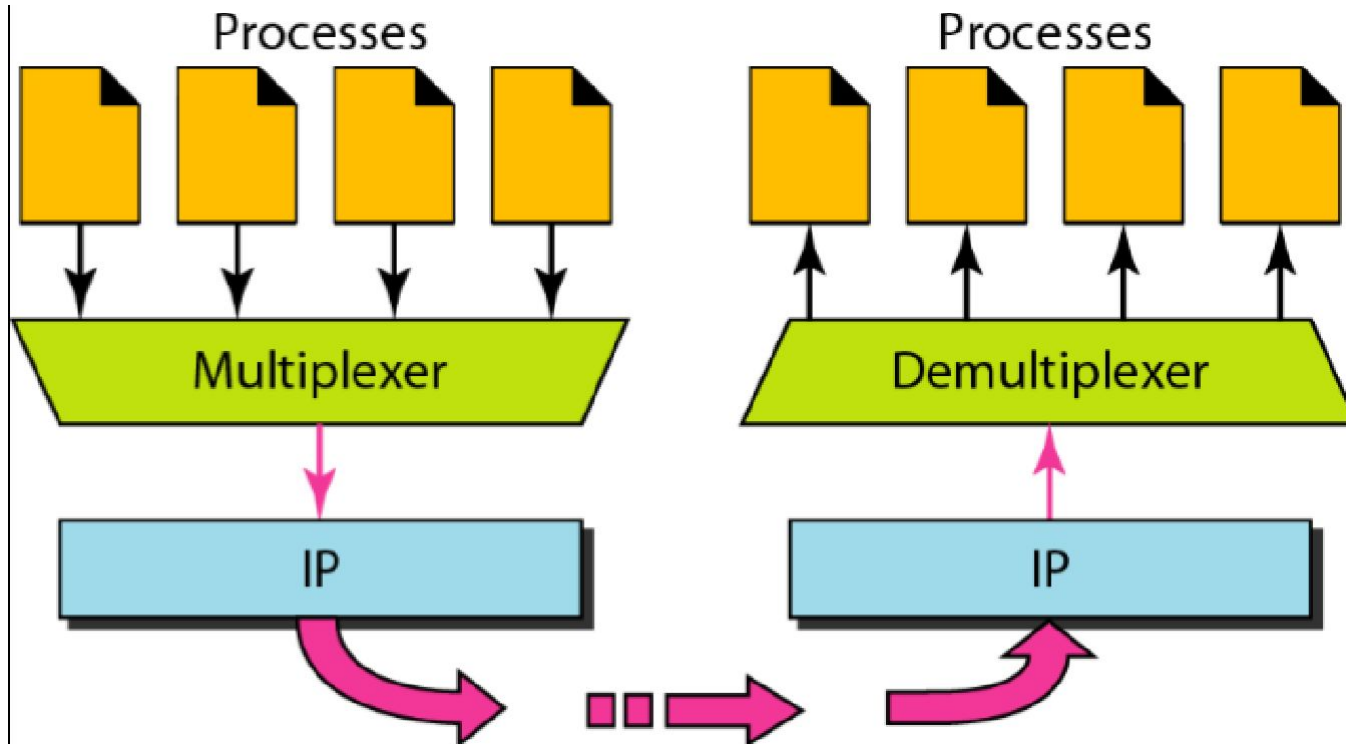
Internet Assigned Number Authority (IANA) Ranges



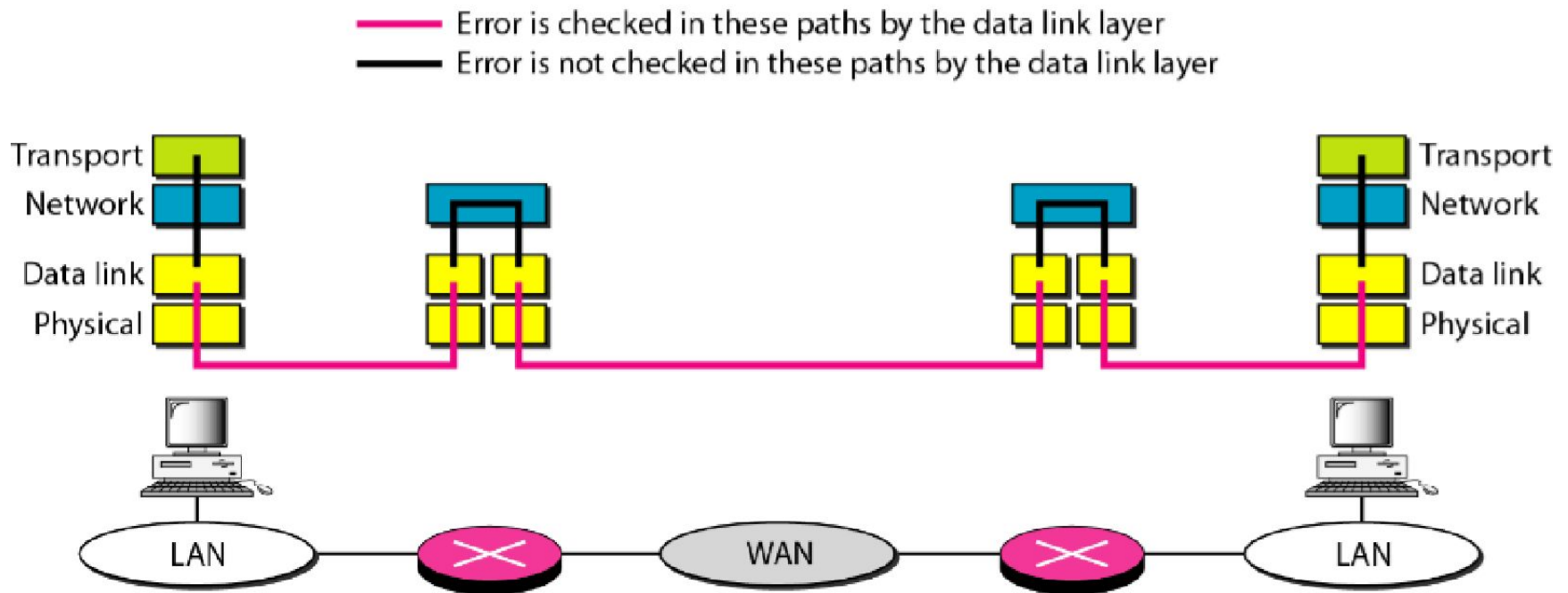
Socket address



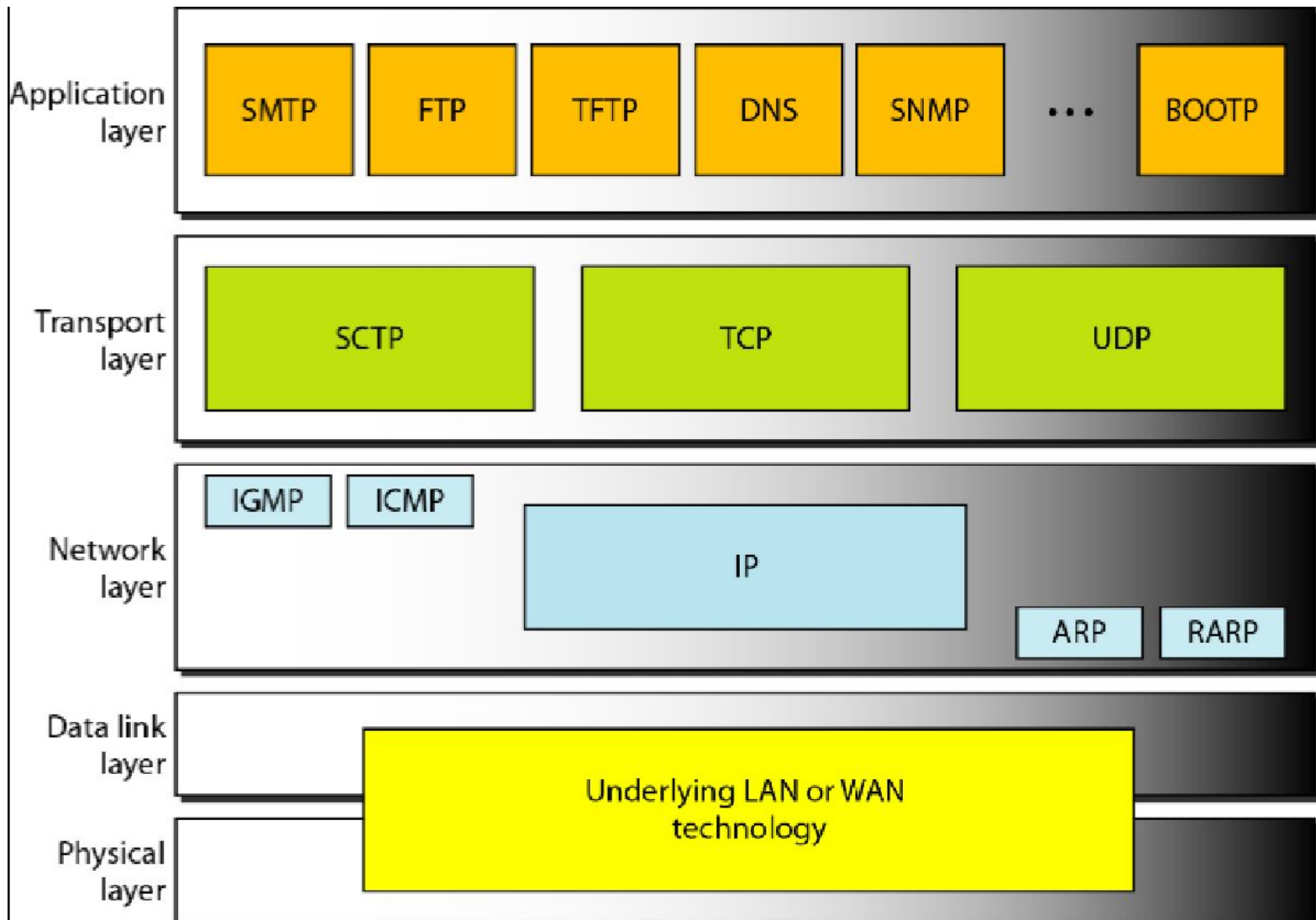
Multiplexing and demultiplexing



Error control



Position of UDP, TCP, and SCTP in TCP/IP



User Datagram Protocol (UDP)

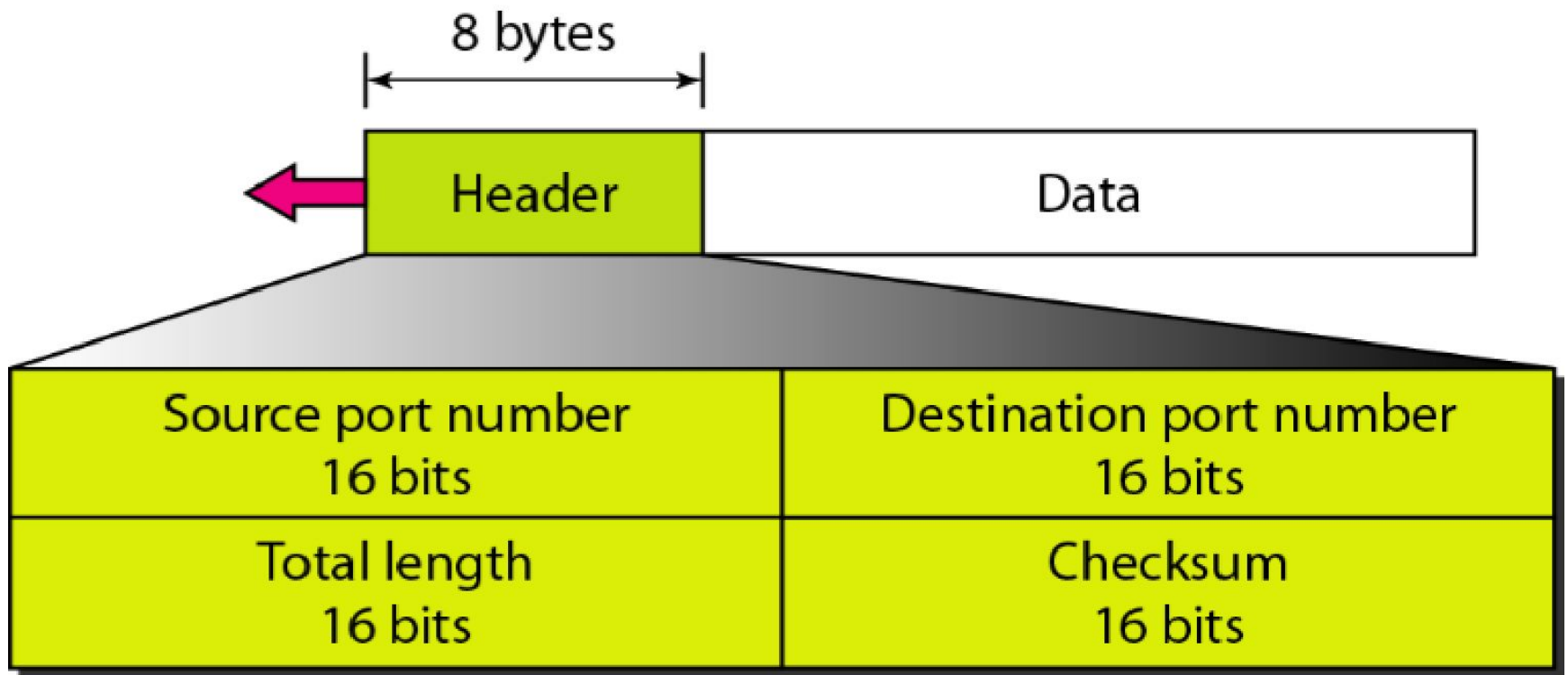
User Datagram Protocol (UDP)

- The User Datagram Protocol (UDP) is called
- a connectionless, unreliable transport
- protocol. It does not add anything to the
- services of IP except to provide process-to process
- communication instead of host-to-host
- communication.

Well-known ports used with UDP

<i>Port</i>	<i>Protocol</i>	<i>Description</i>
7	Echo	Echoes a received datagram back to the sender
9	Discard	Discards any datagram that is received
11	Users	Active users
13	Daytime	Returns the date and the time
17	Quote	Returns a quote of the day
19	Chargen	Returns a string of characters
53	Nameserver	Domain Name Service
67	BOOTPs	Server port to download bootstrap information
68	BOOTPc	Client port to download bootstrap information
69	TFTP	Trivial File Transfer Protocol
111	RPC	Remote Procedure Call
123	NTP	Network Time Protocol
161	SNMP	Simple Network Management Protocol
162	SNMP	Simple Network Management Protocol (trap)

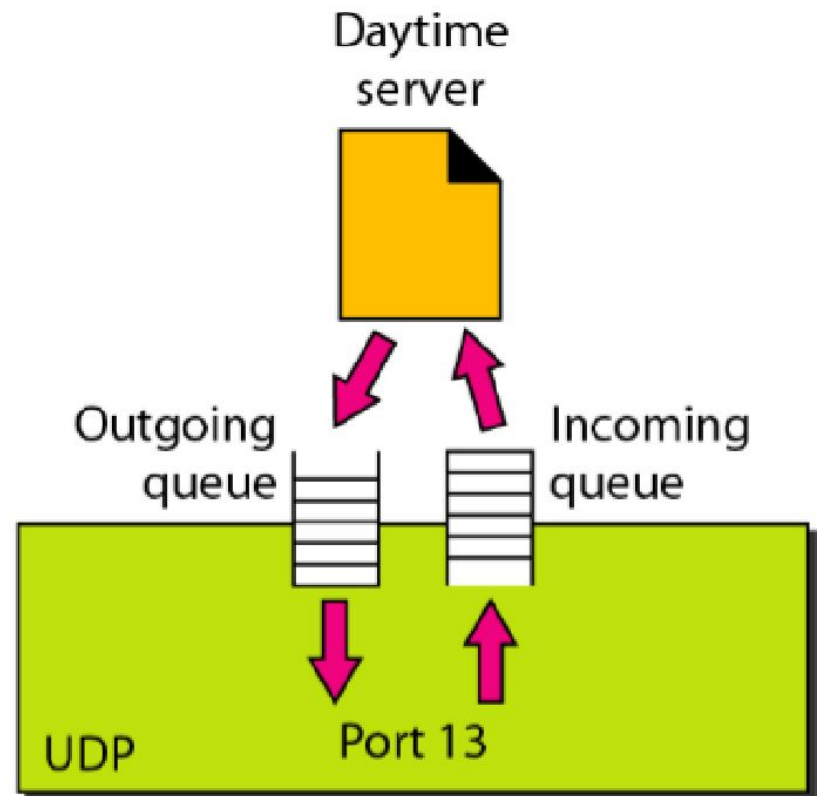
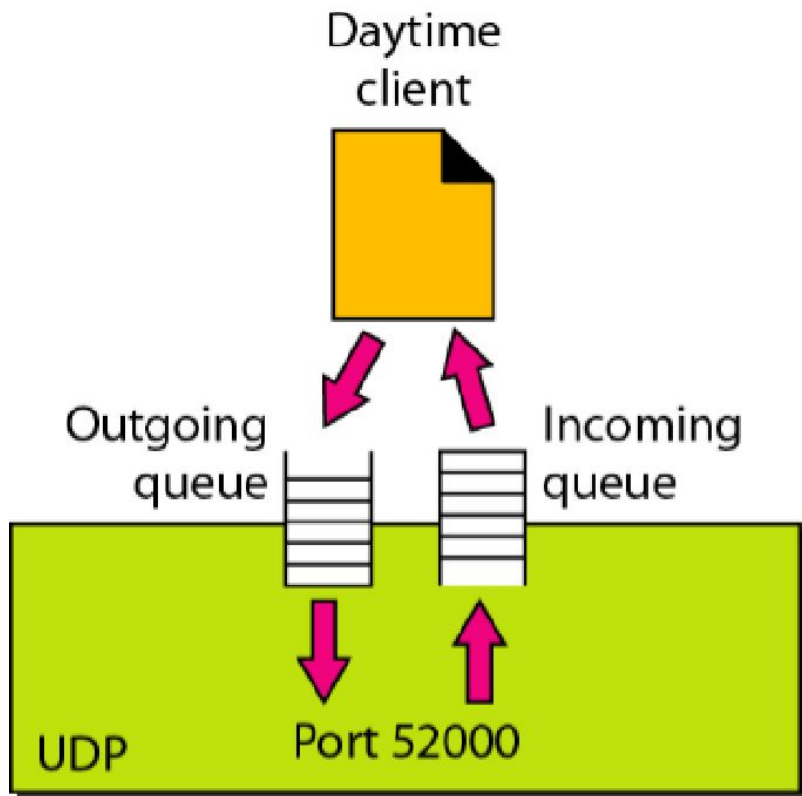
User Datagram Format



UDP Operation

- Connectionless Services
- Flow and Error Control
- No flow control
- No Error Control except Checksum
- Encapsulation & Decapsulation

Queues in UDP



TCP

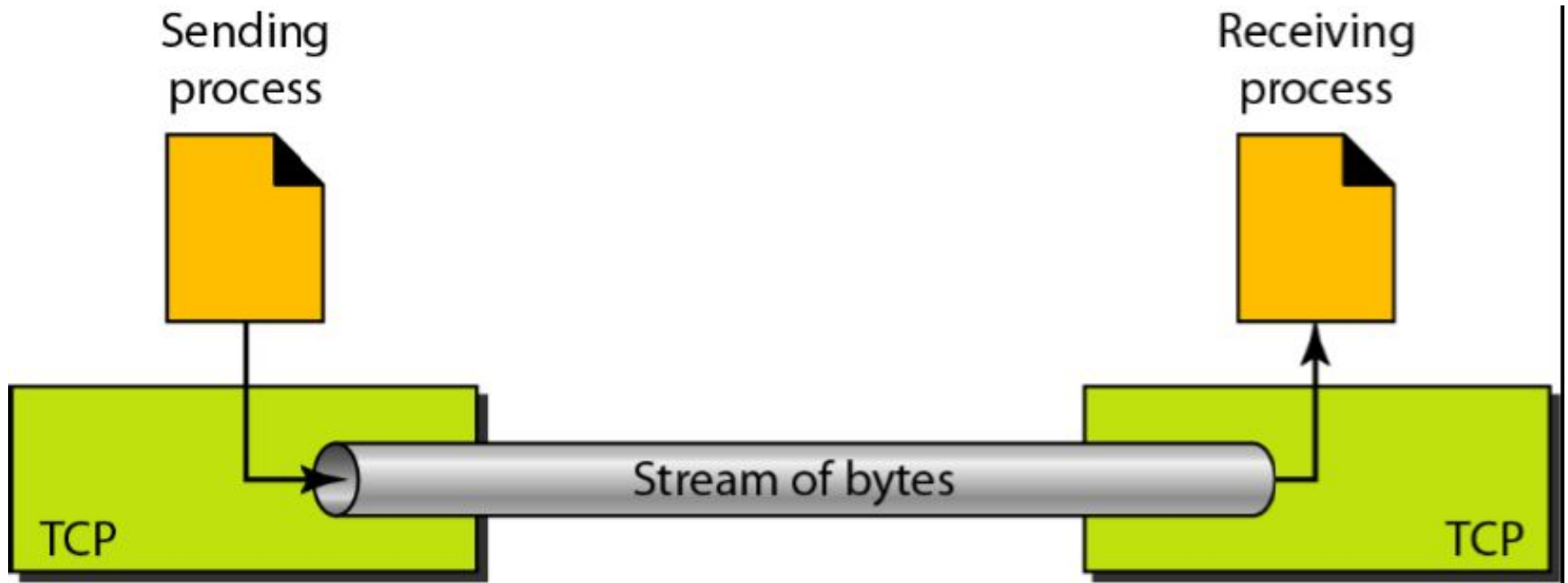
TCP

- TCP is a connection-oriented protocol; it
- creates a virtual connection between two
- TCPs to send data. In addition, TCP uses
- flow and error control mechanisms at the
- transport level.

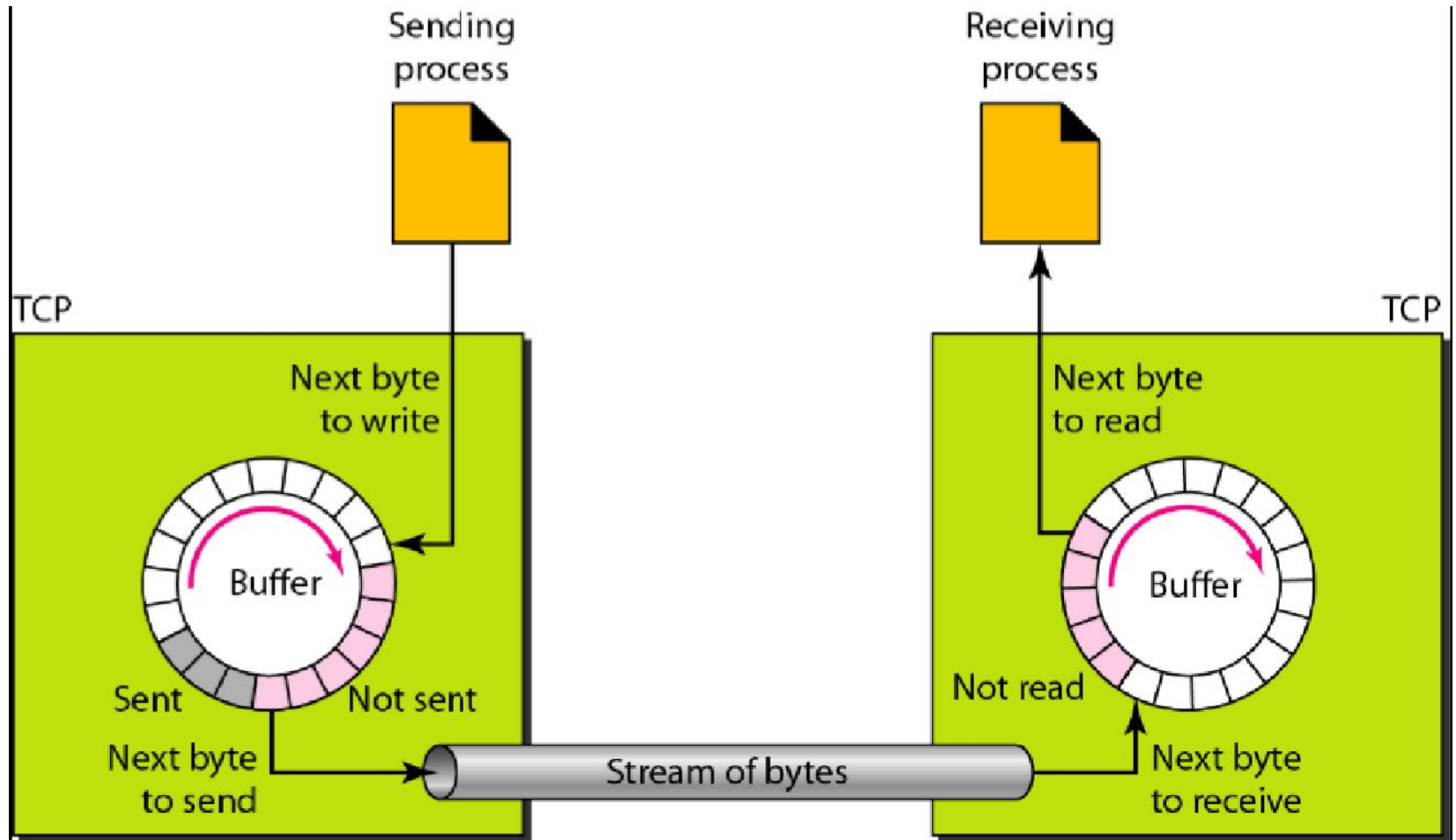
Well-known ports used by TCP

<i>Port</i>	<i>Protocol</i>	<i>Description</i>
7	Echo	Echoes a received datagram back to the sender
9	Discard	Discards any datagram that is received
11	Users	Active users
13	Daytime	Returns the date and the time
17	Quote	Returns a quote of the day
19	Chargen	Returns a string of characters
20	FTP, Data	File Transfer Protocol (data connection)
21	FTP, Control	File Transfer Protocol (control connection)
23	TELNET	Terminal Network
25	SMTP	Simple Mail Transfer Protocol
53	DNS	Domain Name Server
67	BOOTP	Bootstrap Protocol
79	Finger	Finger
80	HTTP	Hypertext Transfer Protocol
111	RPC	Remote Procedure Call

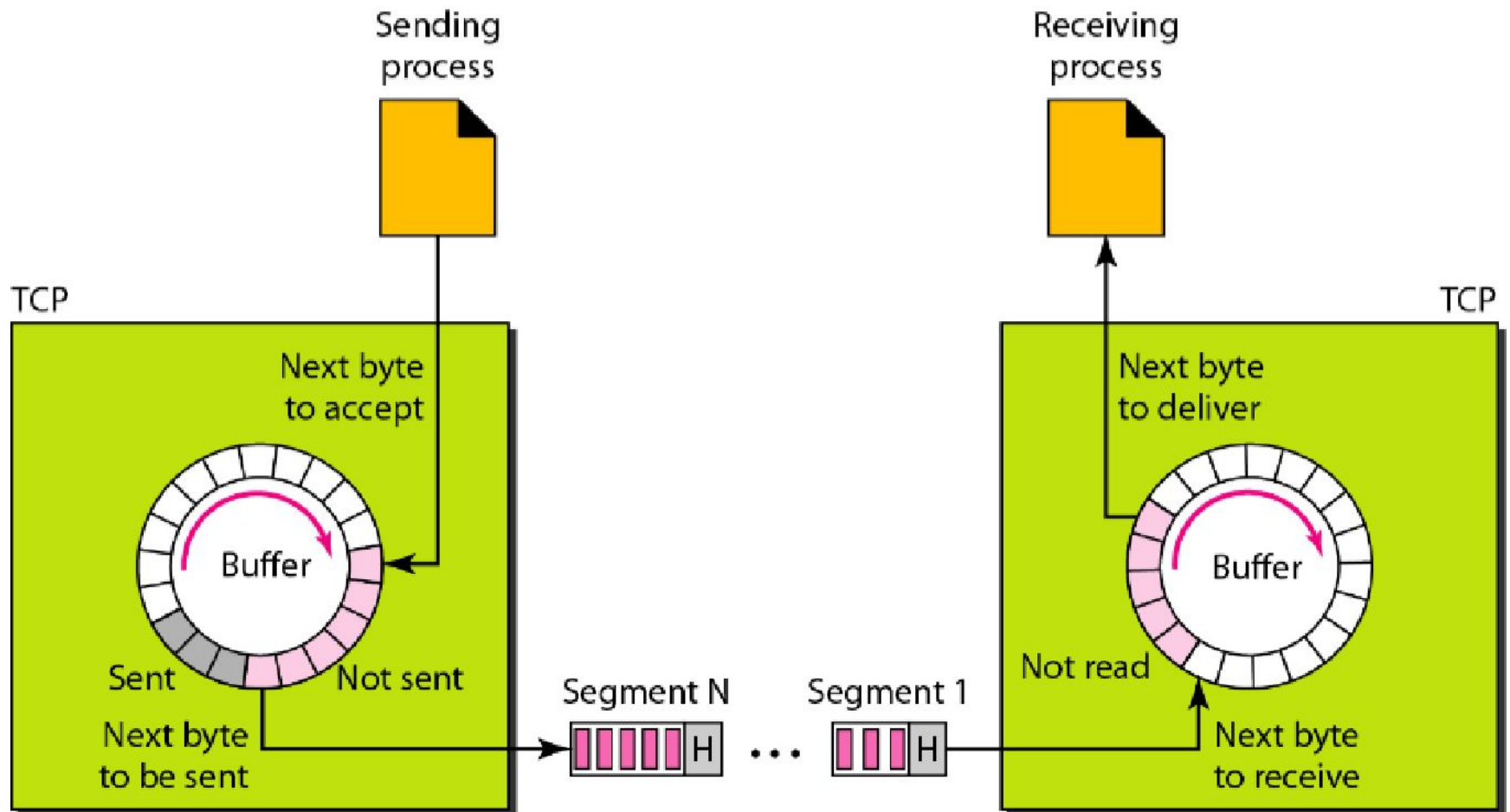
Stream delivery



Sending and receiving



TCP segments



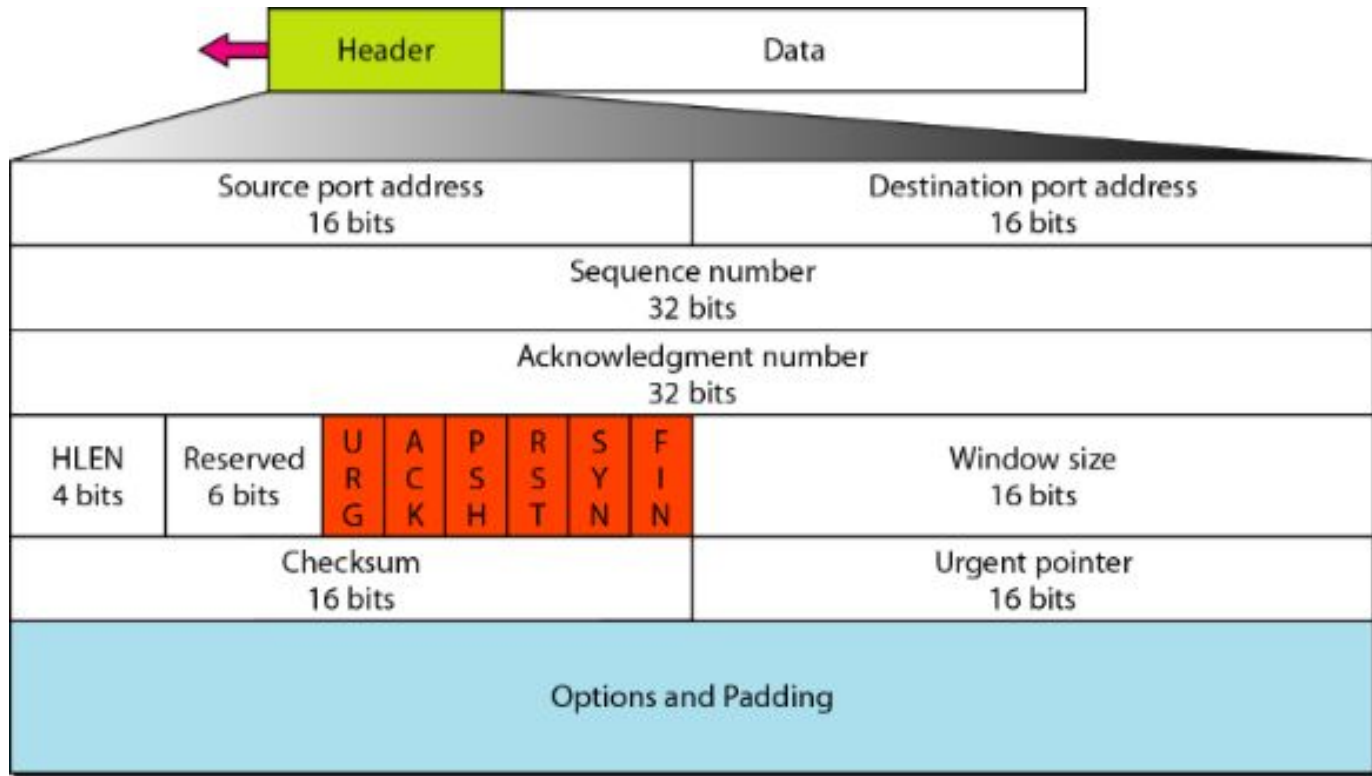
Example

- The bytes of data being transferred in each connection are numbered by TCP. The numbering starts with a randomly generated number.

Segment 1	➡	Sequence Number: 10,001 (range: 10,001 to 11,000)
Segment 2	➡	Sequence Number: 11,001 (range: 11,001 to 12,000)
Segment 3	➡	Sequence Number: 12,001 (range: 12,001 to 13,000)
Segment 4	➡	Sequence Number: 13,001 (range: 13,001 to 14,000)
Segment 5	➡	Sequence Number: 14,001 (range: 14,001 to 15,000)

- The value in the sequence number field of a segment defines the number of the first data byte contained in that segment.
- The value of the acknowledgment field in a segment defines the number of the next byte a party expects to receive. The acknowledgment number is cumulative.

TCP segment format



URG: Urgent pointer is valid

ACK: Acknowledgment is valid

PSH: Request for push

RST: Reset the connection

SYN: Synchronize sequence numbers

FIN: Terminate the connection

URG

ACK

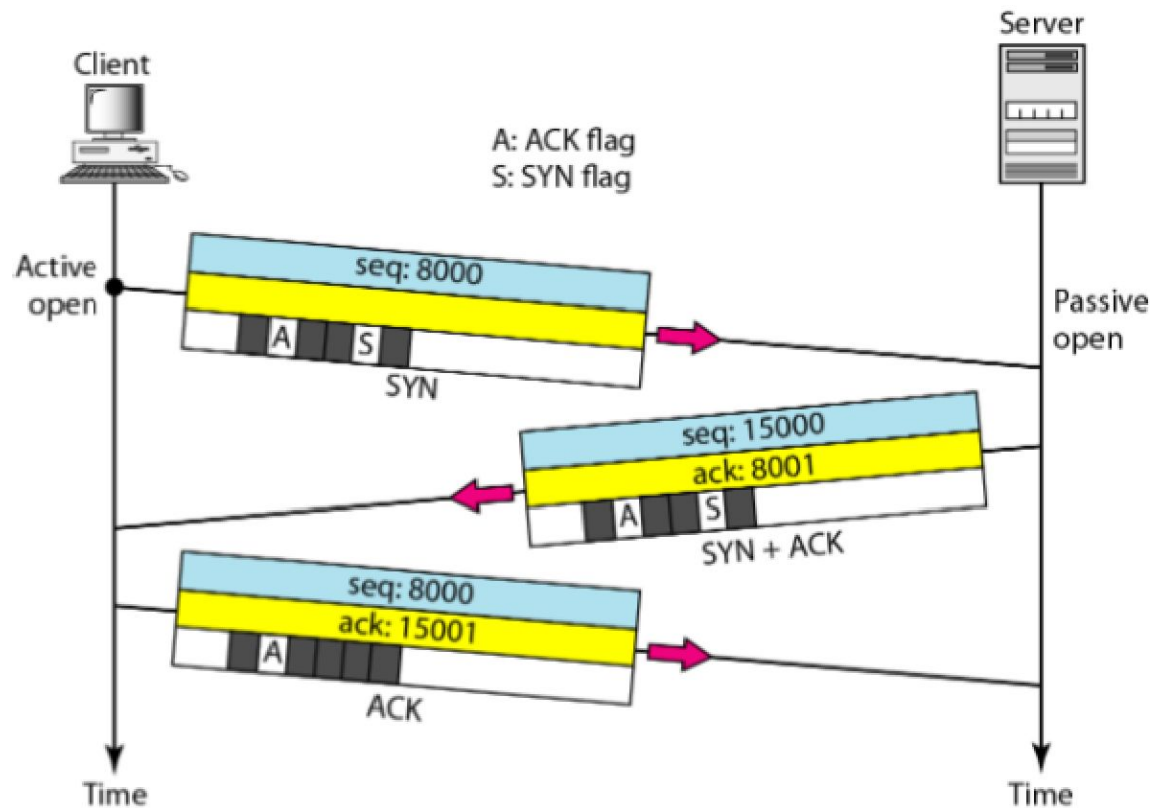
PSH

RST

SYN

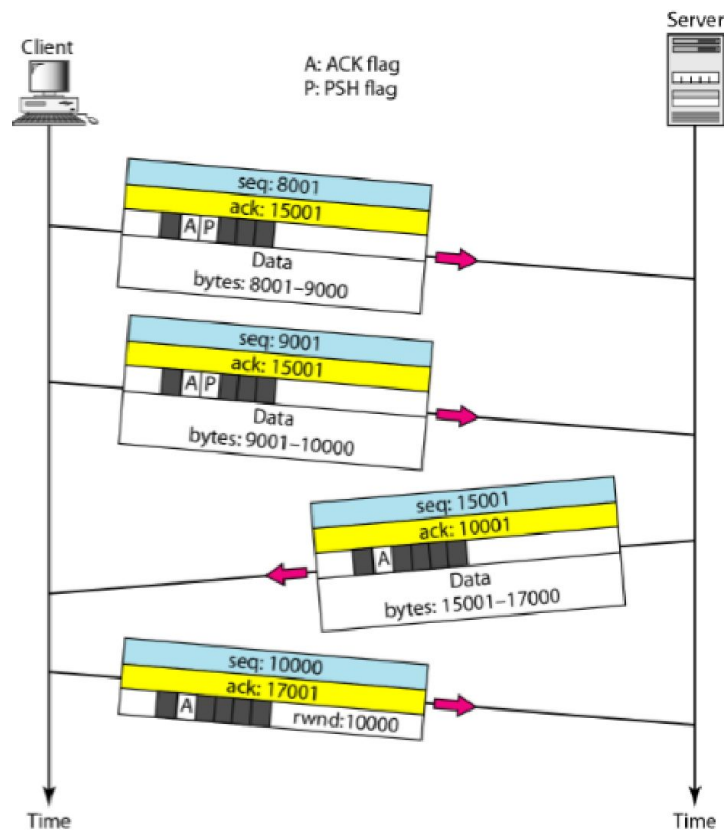
FIN

Connection establishment three-way handshake

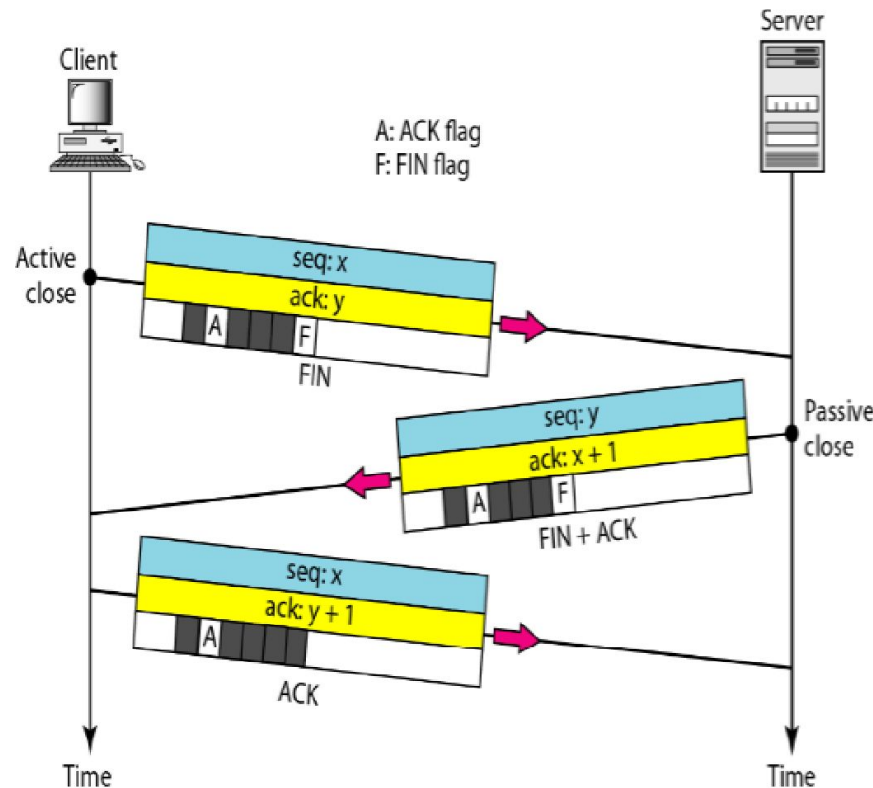


- A SYN segment cannot carry data, but it consumes one sequence number.
- A SYN + ACK segment cannot carry data, but does consume one sequence number.
- An ACK segment, if carrying no data, consumes no sequence number.

Data transfer

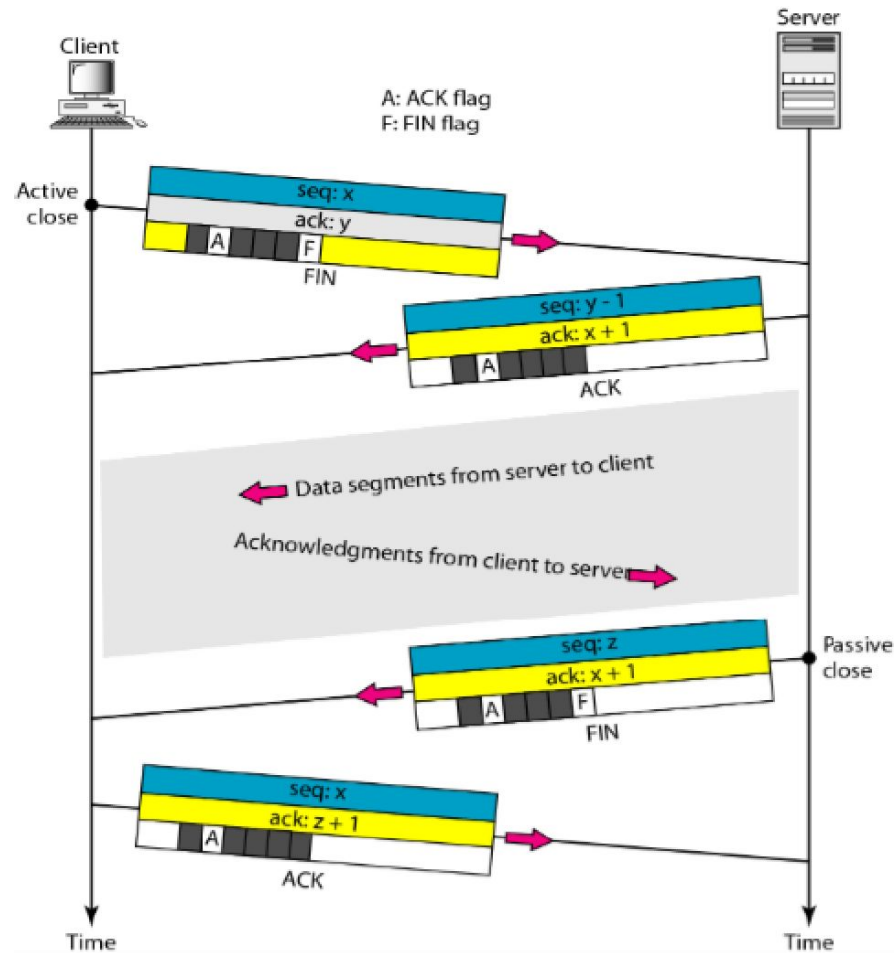


Connection termination using threeway handshaking



- The FIN Segment consumes one Sequence number if it does not carry data.
- The FIN + ACK segment consumes one sequence number if it does not carry data.

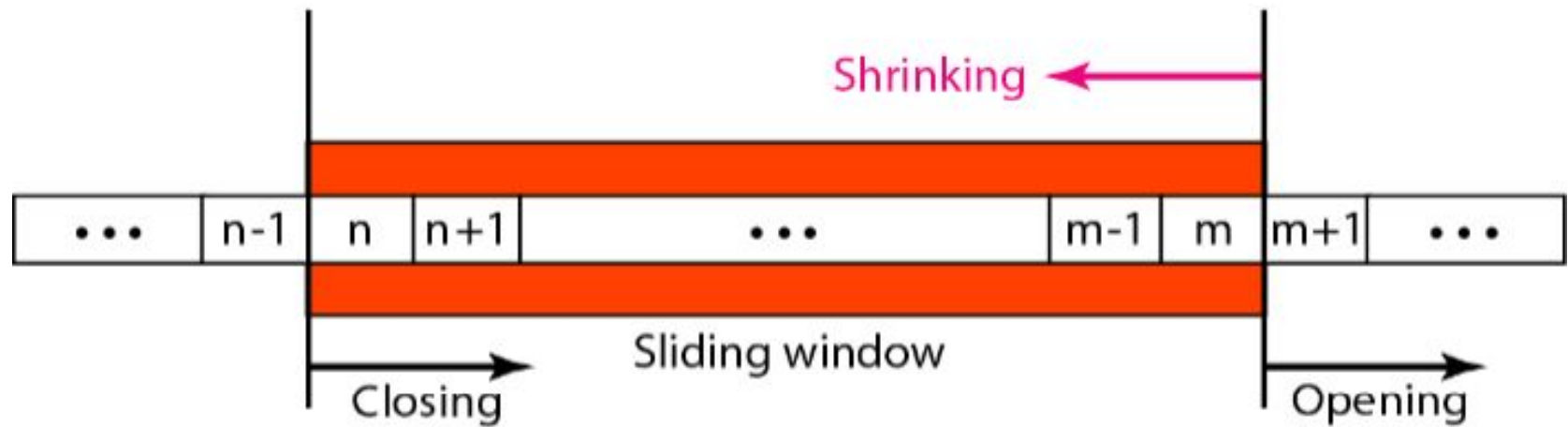
Half Close



Sliding Window Protocol

- A sliding window is used to make transmission more efficient as well as to control the flow of data so that the destination does not become overwhelmed with data.
- TCP sliding windows are byte-oriented.

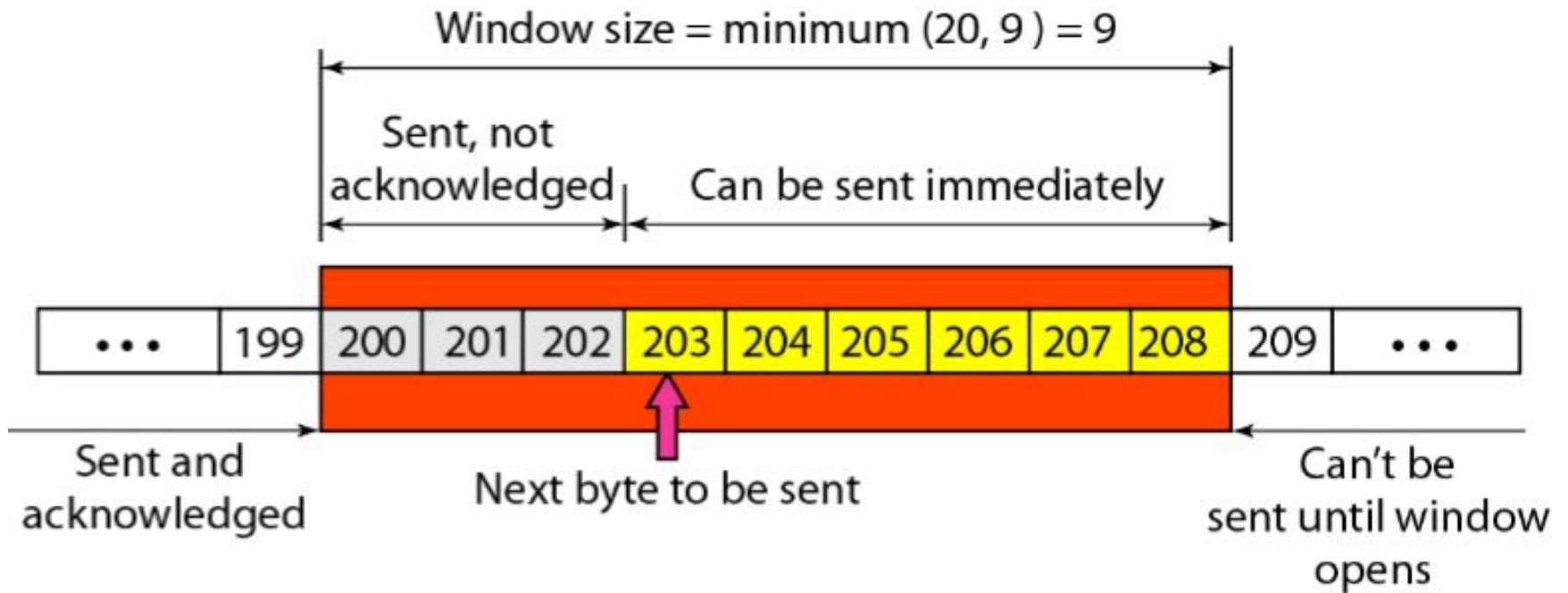
Window size = minimum (rwnd, cwnd)



Example

- *Figure on next slide shows an unrealistic example of a*
- *sliding window. The sender has sent bytes up to 202. We*
- *assume that cwnd is 20 (in reality this value is thousands*
- *of bytes). The receiver has sent an acknowledgment*
- *number of 200 with an rwnd of 9 bytes (in reality this*
- *value is thousands of bytes). The size of the sender*
- *window is the minimum of rwnd and cwnd, or 9 bytes.*
- *Bytes 200 to 202 are sent, but not acknowledged. Bytes*
- *203 to 208 can be sent without worrying about*
- *acknowledgment. Bytes 209 and above cannot be sent.*

Example



NOTE:

- **ACK segments do not consume sequence numbers and are not acknowledged.**

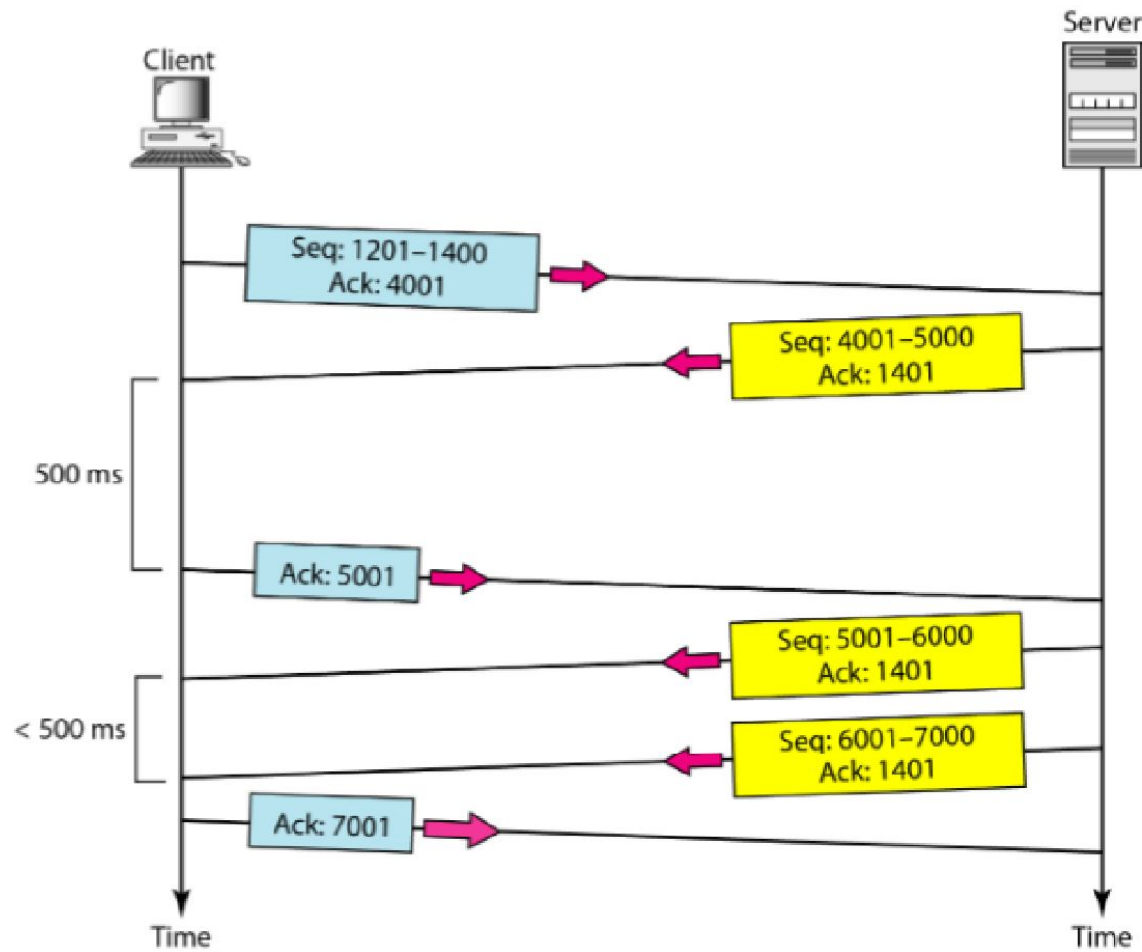
NOTE:

- **In modern implementation ,retransmission occurs if the retransmission timer expires or three duplicate ACK segments have arrived.**
- **No retransmission timer is set for ACK segment**

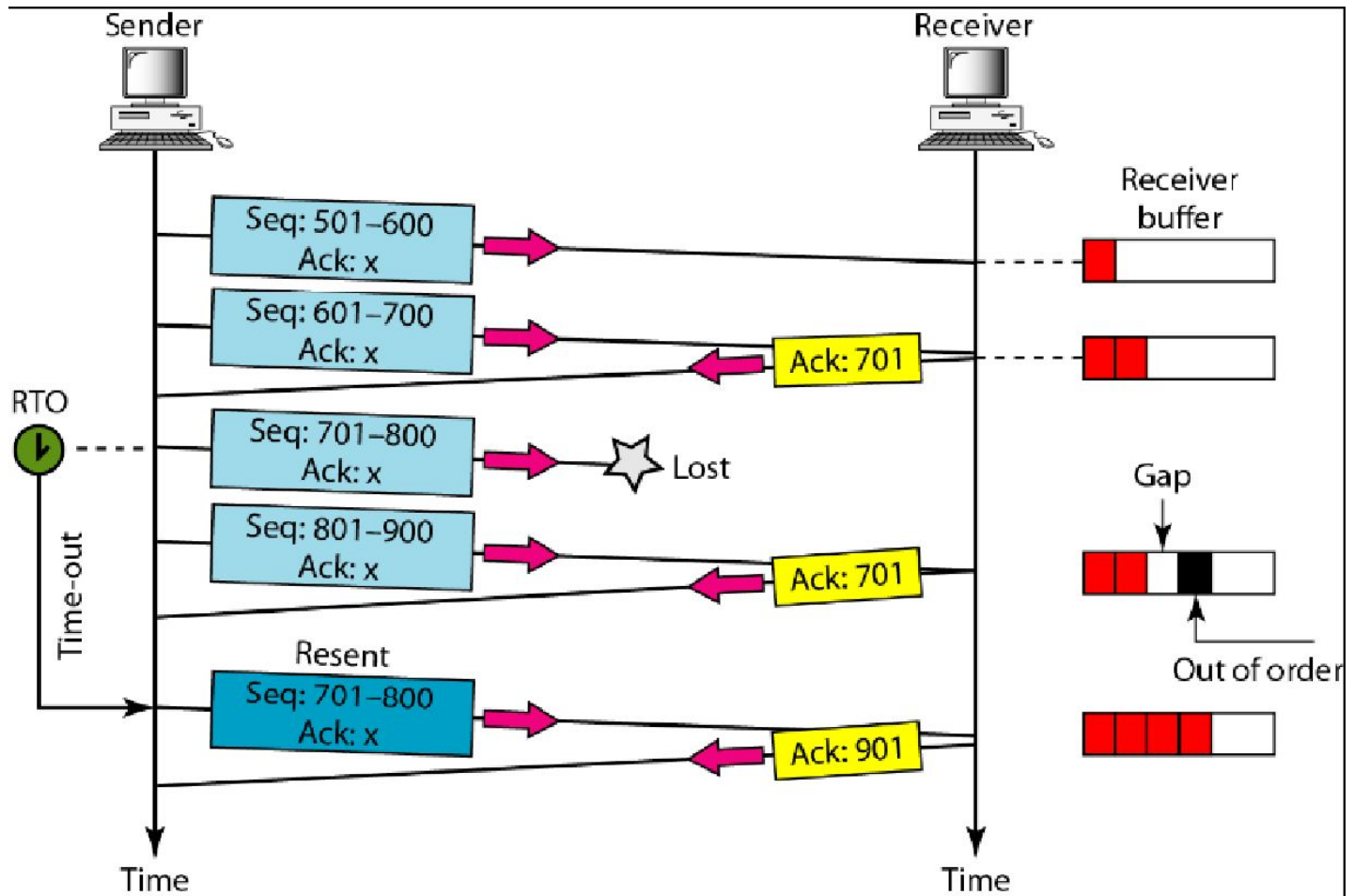
NOTE:

- **Data may arrive out of order be temporarily stored by the receiving TCP, but TCP guarantees that no out-of-order segment is delivered to the process.**

Normal operation



Lost Segment



NOTE:

- **The receiver TCP delivers only ordered data to the process.**

Fast Retransmission

